

Stoner Criterion Evaluation

An operational guide for $(\text{Nb}_{0.8}\text{Ti}_{0.2})_4\text{C}_3$

May 27, 2026

Abstract

This document is an operational guide. It shows step by step *how* the Stoner criterion was evaluated for the 2D MXene $(\text{Nb}_{0.8}\text{Ti}_{0.2})_4\text{C}_3$. For each numerical value, we indicate which Quantum ESPRESSO output file contains it and which formula is applied. The goal is full reproducibility: anyone who runs the same calculations should obtain the same numbers (within $\sim 1\%$ numerical noise).

1 Introduction

1.1 What is the Stoner criterion?

The Stoner criterion is a band-theory test. It predicts when a metal spontaneously becomes ferromagnetic. The condition is:

$$I_{\text{eff}} \cdot N(E_{\text{F}}) > 1 \quad (1)$$

where

- $N(E_{\text{F}})$ is the density of states at the Fermi level (per atom, per spin) computed in the *non-magnetic* state.
- I_{eff} is an effective exchange parameter with units of energy.

If the product exceeds one, the paramagnetic state is unstable against spontaneous spin polarization, and the system becomes ferromagnetic.

1.2 System under study

Composition	$(\text{Nb}_{0.8}\text{Ti}_{0.2})_4\text{C}_3$
Unit cell	56 atoms (6 Ti, 26 Nb, 24 C)
Dimensionality	2D MXene with vacuum
Code	Quantum ESPRESSO 7.4.1
Functional	PBE (GGA)
Hubbard U	$U(\text{Ti-}3d) = 4.3 \text{ eV}$, $U(\text{Nb-}4d) = 3.4 \text{ eV}$

Table 1: Calculation setup.

Pseudopotentials. All pseudopotentials are scalar-relativistic and taken from the SSSP Efficiency library, version 1.3.0:

Element	Mass (a.m.u.)	File	Type
Ti	47.867	ti_pbe_v1.4.uspp.F.UPF	USPP
Nb	92.906	Nb.pbe-spn-kjpaw_psl.0.3.0.UPF	PAW
C	12.011	C.pbe-n-kjpaw_psl.1.0.0.UPF	PAW

Table 2: Pseudopotentials used in the calculations.

2 How I_{eff} is obtained

We cannot extract I_{eff} directly from a DFT calculation. We use the standard approximation: in the ferromagnetic state, the exchange interaction shifts the spin-up and spin-down bands apart by an amount Δ_{ex} , and this shift is proportional to the magnetic moment:

$$\Delta_{\text{ex}} \approx I_{\text{eff}} \cdot m \implies I_{\text{eff}} = \frac{\Delta_{\text{ex}}}{m_{\text{Ti-d}}} \quad (2)$$

Therefore the criterion in Eq. (1) can be rewritten as:

$$\frac{\Delta_{\text{ex}}}{m_{\text{Ti-d}}} \cdot N(E_{\text{F}}) > 1 \quad (3)$$

We need **three numerical inputs**:

- (a) $N(E_{\text{F}})$: from the **non-magnetic** calculation.
- (b) $m_{\text{Ti-d}}$: from the **ferromagnetic** calculation.
- (c) Δ_{ex} : from the **ferromagnetic** calculation.

This is why we run *two* independent self-consistent calculations on the same system.

3 The two calculations we need

3.1 Why two calculations?

The Stoner criterion tests an *instability* of the *paramagnetic* state. Therefore $N(E_{\text{F}})$ must come from the non-magnetic reference; using $N(E_{\text{F}})$ from the spin-polarized result would underestimate it, because the ferromagnetic state has already moved states away from E_{F} .

The local moment $m_{\text{Ti-d}}$ and the band splitting Δ_{ex} , on the other hand, only exist in the ferromagnetic solution.

3.2 Files produced by each calculation

Each calculation is a chain of four Quantum ESPRESSO steps:

$$\text{pw.x (scf)} \rightarrow \text{pw.x (nscf)} \rightarrow \text{dos.x} \rightarrow \text{projwfc.x}$$

The files we will use are summarized in Table 3.

4 Step-by-step procedure

4.1 Step 1: Extract the two Fermi energies

The Fermi energy is printed by `pw.x` at the end of each SCF run. Search for the line:

```
the Fermi energy is X.XXXX ev
```

Calculation	File	Provides
NM (nspin=1)	non-magnetic/outputs/bare.opt	E_F^{NM}
NM	non-magnetic/pdos_files/mxns.pdos_atm# $N(\text{Ti})$ _wfc#4(d)	$\rho^{\text{Ti-}d}(E)$ in NM state
FM (nspin=2)	magnetic/outputs/bare.opt	E_F^{FM} , magnetization
FM	magnetic/pdos/mxns.pdos_atm# $N(\text{Ti})$ _wfc#4(d)	$\rho^{\uparrow, \text{Ti-}d}(E)$, $\rho^{\downarrow, \text{Ti-}d}(E)$

Table 3: Files used in the Stoner evaluation. N runs over the six Ti atoms ($N = 1, \dots, 6$).

Source file

NM: non-magnetic/outputs/bare.opt
FM: magnetic/outputs/bare.opt

In Python this is one line:

```
matches = re.findall(r"the Fermi energy is\s+([-.\\d]+)\s+ev",
                    scf_out, re.IGNORECASE)
E_F = float(matches[-1]) # take the last (converged) value
```

Result

$E_F^{\text{NM}} = 5.0476$ eV
 $E_F^{\text{FM}} = 5.2374$ eV

Convention warning

`dos.x` prints its own Fermi energy in the header of `mxns.dos`. Do **not** use that one it is recomputed with the NSCF smearing and may differ by ~ 10 meV. Always use the value from the SCF (`bare.opt`).

4.2 Step 2: Compute $N(E_F)$ from the non-magnetic PDOS

The non-magnetic calculation produces, for each Ti atom, a file named `mxns.pdos_atm# $N(\text{Ti})$ _wfc#4(d)` with seven columns. The format is described in Table 4.

Column	Quantity
1	energy E (eV)
2	total Ti- d LDOS (per spin)
3	PDOS d_{z^2}
4	PDOS d_{xz}
5	PDOS d_{yz}
6	PDOS $d_{x^2-y^2}$
7	PDOS d_{xy}

Table 4: Column layout of the NM PDOS files (nspin=1, 7 columns).

What to do. We sum the LDOS over all six Ti atoms, divide by six (to get a per-atom quantity), and evaluate at $E = E_F^{\text{NM}}$:

$$N(E_F) = \frac{1}{6} \sum_{N=1}^6 \rho_N^{\text{Ti-}d, \text{NM}}(E_F^{\text{NM}}). \quad (4)$$

Convention warning

In `nspin=1`, the LDOS that Quantum ESPRESSO writes is *already per spin channel*. The integral $\int_{-\infty}^{E_F} \text{LDOS}(E) dE$ gives $N_{\text{total}}/2$, not N_{total} . For the Stoner formula, the per-spin convention is exactly what we want so we do **not** multiply by 2.

Pseudo-code.

```
ldos_d_NM = sum(data_Ti[i][:, 1] for i in range(6))
ldos_d_NM_per_Ti = ldos_d_NM / 6
N_EF = numpy.interp(E_F_NM, E_grid, ldos_d_NM_per_Ti)
```

Result

$N(E_F) = 1.1920 \text{ states/eV/Ti/spin}$

4.3 Step 3: Compute $m_{\text{Ti-d}}$ from the ferromagnetic PDOS

The ferromagnetic PDOS files have 13 columns (spin-resolved): see Table 5.

Column	Quantity
1	energy E (eV)
2, 3	total Ti-d LDOS $\uparrow, \downarrow$
4, 5	PDOS d_{z^2} $\uparrow, \downarrow$
6, 7	PDOS d_{xz} $\uparrow, \downarrow$
8, 9	PDOS d_{yz} $\uparrow, \downarrow$
10, 11	PDOS $d_{x^2-y^2}$ $\uparrow, \downarrow$
12, 13	PDOS d_{xy} $\uparrow, \downarrow$

Table 5: Column layout of the FM PDOS files (`nspin=2`, 13 columns).

What to do. Sum the $\text{LDOS}^\uparrow$ and $\text{LDOS}^\downarrow$ over the six Ti atoms, divide by six, integrate from the lower bound to E_F^{FM} , and take the difference:

$$N_{\text{Ti-d}}^\uparrow = \frac{1}{6} \sum_{N=1}^6 \int_{-\infty}^{E_F^{\text{FM}}} \rho_{N,\text{Ti-d}}^\uparrow(E) dE, \quad (5)$$

$$N_{\text{Ti-d}}^\downarrow = \frac{1}{6} \sum_{N=1}^6 \int_{-\infty}^{E_F^{\text{FM}}} \rho_{N,\text{Ti-d}}^\downarrow(E) dE, \quad (6)$$

$$m_{\text{Ti-d}} = N_{\text{Ti-d}}^\uparrow - N_{\text{Ti-d}}^\downarrow. \quad (7)$$

The integral has units of $(\text{states/eV}) \times \text{eV} = \text{electrons}$. Each unbalanced electron carries one Bohr magneton, so the difference is $m_{\text{Ti-d}}$ in μ_B/Ti .

Pseudo-code.

```
ldos_up_FM = sum(data_Ti_FM[i][:, 1] for i in range(6)) / 6
ldos_dn_FM = sum(data_Ti_FM[i][:, 2] for i in range(6)) / 6
mask = (E_grid_FM <= E_F_FM)
N_up = numpy.trapezoid(ldos_up_FM[mask], E_grid_FM[mask])
N_dn = numpy.trapezoid(ldos_dn_FM[mask], E_grid_FM[mask])
m_Ti_d = N_up - N_dn
```

Result

$$\begin{aligned} N_{\text{Ti-}d}^{\uparrow} &= 1.6103 \text{ e/Ti} \\ N_{\text{Ti-}d}^{\downarrow} &= 0.6951 \text{ e/Ti} \\ m_{\text{Ti-}d} &= 0.9152 \mu_{\text{B}}/\text{Ti} \end{aligned}$$

Cross-check. Quantum ESPRESSO also writes Löwdin moments directly inside the file `magnetic/outputs/p` in lines like

```
Atom # 1: ... spin up = 5.9450, ..., d = 1.6130, dz2= 0.1199, ...
           spin down = 4.9599, ..., d = 0.6996, dz2= 0.0912, ...
```

Subtracting $d^{\downarrow}$ from $d^{\uparrow}$ for each Ti and averaging gives an independent estimate of $m_{\text{Ti-}d}$ that should agree with the PDOS integral to within $\sim 1\%$. This is a useful sanity check.

4.4 Step 4: Compute the exchange splitting Δ_{ex}

We use the **center-of-mass method**: Δ_{ex} is the energy difference between the centers of mass of the spin-up and spin-down Ti- d bands.

Formulas. Let $W = [W_1, W_2]$ be an energy window (relative to E_{F}^{FM}). Define:

$$\varepsilon_d^{\uparrow} = \frac{\int_W E \rho_{\text{Ti-}d}^{\uparrow}(E) dE}{\int_W \rho_{\text{Ti-}d}^{\uparrow}(E) dE}, \quad (8)$$

$$\varepsilon_d^{\downarrow} = \frac{\int_W E \rho_{\text{Ti-}d}^{\downarrow}(E) dE}{\int_W \rho_{\text{Ti-}d}^{\downarrow}(E) dE}, \quad (9)$$

$$\Delta_{\text{ex}} = \varepsilon_d^{\downarrow} - \varepsilon_d^{\uparrow}. \quad (10)$$

Window choice. We use $W = [-8, +10]$ eV (relative to E_{F}). This is wide enough to contain the entire Ti- d band on both spin channels. The robustness to this choice is checked below.

Pseudo-code.

```
E_rel = E_grid_FM - E_F_FM
mask = (E_rel >= -8.0) & (E_rel <= 10.0)
eps_up = (numpy.trapezoid(E_rel[mask] * ldos_up_FM[mask], E_rel[mask])
         / numpy.trapezoid(ldos_up_FM[mask], E_rel[mask]))
eps_dn = (numpy.trapezoid(E_rel[mask] * ldos_dn_FM[mask], E_rel[mask])
         / numpy.trapezoid(ldos_dn_FM[mask], E_rel[mask]))
Delta_ex = eps_dn - eps_up
```

Result

$$\begin{aligned} \varepsilon_d^{\uparrow} &= +0.6173 \text{ eV} \\ \varepsilon_d^{\downarrow} &= +1.7844 \text{ eV} \\ \Delta_{\text{ex}} &= 1.1672 \text{ eV} \end{aligned}$$

4.5 Step 5: Combine the pieces

We now have everything we need. Following Eq. (2):

$$I_{\text{eff}} = \frac{\Delta_{\text{ex}}}{m_{\text{Ti-}d}} = \frac{1.1672 \text{ eV}}{0.9152 \mu_{\text{B}}} = 1.2753 \text{ eV}. \quad (11)$$

Applying the Stoner criterion of Eq. (1):

$$I_{\text{eff}} \cdot N(E_F) = 1.2753 \times 1.1920 = 1.5201 \quad (12)$$

Result

Stoner criterion: **SATISFIED**

$I_{\text{eff}} \cdot N(E_F) = 1.5201 > 1$, with a margin of +52.0%.

The non-magnetic state is unstable. The system favours an itinerant Stoner-type ferromagnetic ordering on the Ti sublattice.

5 Robustness checks

5.1 Sensitivity to the window in Δ_{ex}

The center-of-mass formula in Eqs. (8)–(10) depends on the chosen energy window. We have verified that the result is essentially insensitive to this choice:

Window (eV)	Δ_{ex} (eV)	Change vs. default
$[-8, +10]$ (default)	1.1672	–
$[-7, +8]$	1.1672	0.00%
$[-6, +6]$	1.1646	–0.22%
$[-10, +10]$	1.1671	–0.01%

Table 6: Exchange splitting under different integration windows.

5.2 Cross-check between methods

The Ti- d moment computed by PDOS integration ($m_{\text{Ti-}d} = 0.9152 \mu_B/\text{Ti}$) agrees with the Löwdin moment in `magnetic/outputs/pdos.opt` to within $\sim 0.4\%$. This is the expected level of internal consistency.

6 Physical interpretation

- The product $I_{\text{eff}} \cdot N(E_F) = 1.52$ is well above the critical value of 1. The system is unambiguously above threshold.
- The effective exchange $I_{\text{eff}} \approx 1.28$ eV is in the range typical of early 3d transition metals.
- The bandsplitting $\Delta_{\text{ex}} \approx 1.17$ eV is comparable to that of bulk iron (~ 1.5 eV) and confirms that the Ti- d shell is strongly exchange-split.
- Because the criterion is satisfied with a +52% margin, the ferromagnetic alignment of the Ti sublattice is robust to small variations in calculation parameters (smearing, k -grid, Hubbard U).

7 What this criterion does NOT prove

The Stoner criterion answers *one specific question*: whether the paramagnetic state is unstable against ferromagnetic alignment on the relevant orbital. It does **not** prove the following:

- **That FM is the ground state.** An antiferromagnetic configuration could still be lower in energy. To rule that out, we would need to compute candidate AFM configurations explicitly.
- **The Curie temperature.** A mean-field Stoner estimate systematically overestimates T_c . A more reliable value requires exchange constants J_{ij} (e.g., from Wannier functions) plus a Monte Carlo or magnon spectrum calculation.
- **Half-metallicity.** The system is not half-metallic: the spin polarization at E_F on the total DOS is only $\approx 2.4\%$, with both spin channels having finite DOS.
- **Dynamic stability.** The phonon spectrum has not been computed in the present work.

8 Summary

Verdict: Stoner criterion satisfied with a margin of +52.0%. See the full table of inputs, outputs, and intermediate quantities below.

Quantity	Value	Source
E_F^{NM}	5.0476 eV	non-magnetic/outputs/bare.opt
E_F^{FM}	5.2374 eV	magnetic/outputs/bare.opt
$N(E_F)$	1.1920 states/eV/Ti/spin	NM PDOS Ti- d , interpolated at E_F^{NM}
$N_{\text{Ti-}d}^{\uparrow}$	1.6103 e/Ti	FM PDOS, $\int_{-\infty}^{E_F^{\text{FM}}} \rho^{\uparrow}$
$N_{\text{Ti-}d}^{\downarrow}$	0.6951 e/Ti	FM PDOS, $\int_{-\infty}^{E_F^{\text{FM}}} \rho^{\downarrow}$
$m_{\text{Ti-}d}$	0.9152 μ_B/Ti	$N_{\text{Ti-}d}^{\uparrow} - N_{\text{Ti-}d}^{\downarrow}$
Δ_{ex}	1.1672 eV	center-of-mass, window $[-8, +10]$ eV
I_{eff}	1.2753 eV	$\Delta_{\text{ex}}/m_{\text{Ti-}d}$
$I_{\text{eff}} \cdot N(E_F)$	1.5201	Stoner product

Table 7: Summary of all quantities used and obtained in the Stoner evaluation.